



University of Pisa · MSc Computer Engineering

Plant Disease Detection and Classification

A field-image diagnosis system: locate the lesion, name the disease, and test how well it holds up on real photos.

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The Problem

Turn a photo of a diseased leaf into a computer-friendly diagnosis.

1 Instant field diagnosis

A farmer or gardener points a phone at a leaf and gets an answer on the spot.

2 Early outbreak flagging

Catch a disease before it spreads across the whole field.

3 A building block

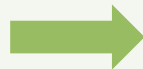
Drop it into a larger crop-monitoring or advisory assistant.

INPUT

Leaf photograph

OUTPUT

Disease + confidence + lesion region



Real field photo, varied lighting, angle and background make this harder than a clean lab image.

Two Approaches We Compared

Both are built on the same dataset, the comparison is the point.

APPROACH A

Standalone, one model

A single YOLO model localizes the lesion and directly assigns one of 115 disease classes in one pass.

Leaf image → **disease-aware YOLO** → disease name

Finds AND names in the same step, one hard joint task.

APPROACH B

Two-stage pipeline

A class-agnostic YOLO first finds the lesion and boxes it, no name. We crop that region and a separate classifier names the disease.

lesion YOLO → **crop** → **classifier** → **name**

The classifier only sees the lesion, not the background, so it can't lean on background cues.

The Dataset, PlantSeg

7,774 real field photographs, each with a lesion mask and a disease label.

Pixel-level mask

Binary lesion mask per image, we derive bounding boxes from it.

115 disease classes

Across 34 plant species.

Official splits

Train / validation / test provided with the dataset.

Partition	Images	Classes present
Training	5,367	115
Validation	846	114
Test	1,561	114

Median 55 images per class

Smallest class: 3 images · largest: 323, a long-tailed, imbalanced set.

Why masks matter

Masks and metadata are the single source of truth. We derive every box from the mask rather than trusting any pre-supplied annotation, so localization and the Grad-CAM check share the same ground truth.



Computer Vision Models

Localization, classification, and an explanation check.

LOCALIZATION

YOLO11n, class-agnostic

One "lesion" class. Just finds where the disease is.

YOLO11n, disease-aware

115 classes. Finds and names in one shot (Approach A).

CLASSIFICATION

Small CNN from scratch

Baseline, no pretraining.

EfficientNetB0

ImageNet-pretrained, fine-tuned.

MobileNetV3-Large

ImageNet-pretrained, fine-tuned.

EXPLANATION

Grad-CAM

Post-hoc on the trained classifier, not trained separately. Shows which pixels drove the prediction, scored against the real mask.

Training Setup

Consistent settings across every experiment.

YOLO epochs	50
Input size	640 ² detect · 224 ² classify
Batch size	16
Classes	115
Optimizer	YOLO auto · AdamW (classifiers)
Device	Apple M1 Pro (MPS)
Seed	42

Selected on macro F1

Not raw accuracy.

Class sizes are unbalanced. Some diseases have only a handful of images. Macro F1 weights every class equally regardless of how many examples it has, so a model can't win by only nailing the common diseases.



5-Fold Cross-Validation

Measuring training stability, not final test performance.

- 1 Partition** The 5,367-image training split, divided into 5 stratified folds by disease class.
- 2 Rare classes** Two classes with only 2 and 4 images are too small to stratify, distributed round-robin instead.
- 3 Per fold** Train on 4 folds, validate on the held-out one. Every model (2 YOLO + 3 classifiers) retrained from scratch.
- 4 Scale** 25 independent training runs, 50 epochs each.

The official validation and test splits are never touched here.

Cross-validation tells us the results are stable across different partitions, the final numbers on later slides come from the untouched held-out test set.



Lesion Detection Results

Final checkpoints on the held-out test split (1,561 images).

Model	Precision	Recall	mAP@50	mAP@50–95
Class-agnostic lesion YOLO	85.3%	81.9%	88.2%	61.5%
Disease-aware standalone YOLO	54.9%	49.9%	50.2%	34.5%

Why the big gap?

The class-agnostic detector only has to find a lesion. The standalone detector must find AND correctly name it in the same step, a much harder joint task. This is the core argument for splitting the problem into two stages.



Lesion detector output, box + confidence on a real field image.

End-to-End Classification Results

Full system, held-out test split.

System	Accuracy	Macro F1	Bal. acc.	Coverage
Standalone disease-aware YOLO	61.4%	53.3%	49.6%	86.5%
Lesion YOLO + baseline CNN	41.3%	29.0%	30.0%	100%
Lesion YOLO + EfficientNetB0	66.7%	59.0%	58.7%	100%
Lesion YOLO + MobileNetV3-Large	66.8%	58.2%	58.7%	100%

Coverage = share of images the system returns an answer for. The two-stage pipeline falls back to the full image when no lesion is found, this happened for just 38 of 1,561 images (2.4%), so coverage stays 100%.

+5 points

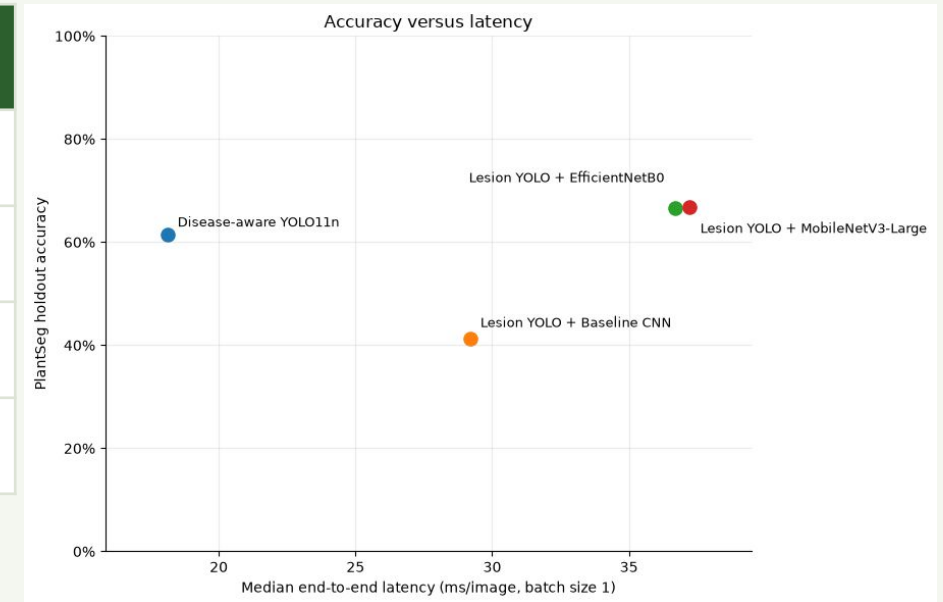
The two-stage pipeline beats the single-shot standalone YOLO by about 5 points of accuracy.

*Cost: two models instead of one.
EfficientNetB0 and MobileNetV3 land within a whisker of each other.*

Accuracy vs. Latency

Is the two-stage accuracy gain worth running two models? Timed at batch size 1 on the training hardware.

System	Median (ms)	P95 (ms)	Throughput (img/s)	Params	Size
Standalone disease-aware YOLO	18.1	231.3	26.3	2.66M	5.6 MB
Lesion YOLO + baseline CNN	29.2	65.0	28.7	3.01M	10.5 MB
Lesion YOLO + EfficientNetB0	36.7	66.5	24.5	6.74M	55.8 MB
Lesion YOLO + MobileNetV3-Large	37.2	64.1	24.3	6.94M	58.0 MB



Speed doesn't track accuracy. The fastest system (baseline CNN) is also the least accurate.

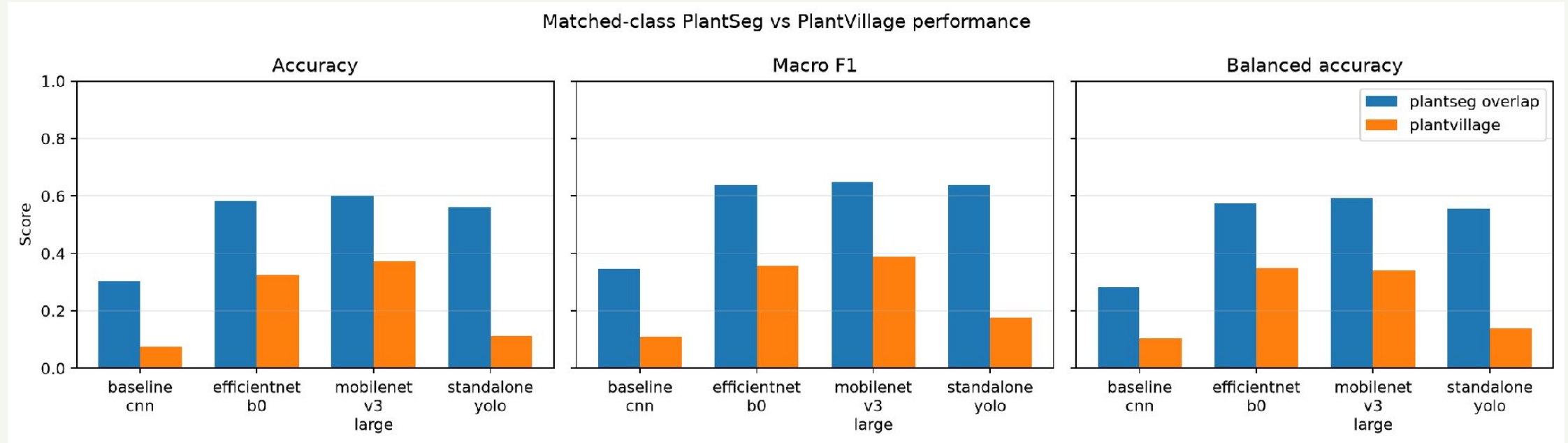
MobileNetV3 trades a little throughput for +29 points of macro F1 over it.

Accuracy vs. median latency, one point per system.

Standalone YOLO has the best median and smallest checkpoint, but its P95 is far worse: some images take much longer. All four are fast enough for a single-image advisory tool.

Cross-Domain Evaluation

Trained on PlantSeg field photos, tested on PlantVillage studio photos. 21 shared classes, no fine-tuning.



Classifier pipeline	PlantSeg macro F1	PlantVillage macro F1	Change
Baseline CNN	34.5%	10.9%	-23.6pp
EfficientNetB0	63.6%	35.7%	-27.9pp
MobileNetV3-Large	64.8%	39.0%	-25.7pp
Standalone disease-aware YOLO -	63.9%	17.6%	-46.3pp

No system generalizes cleanly. Standalone YOLO drops most: its coverage falls from 86.8% to 43.2%, it finds no lesion on clean single-leaf photos.

Grad-CAM vs Ground-Truth Masks

Is the classifier looking at the disease, or the background?

CORRECT: attention lands on the rot · panels: photo / true mask / Grad-CAM



24.2%

IoU at the top-20% activation quantile

41.5%

of Grad-CAM energy falls inside the mask, which covers only ~21% of the image on average

FAILURE: wrong class, attention off-target



58.8%

pointing-game hit rate: hottest pixel lands inside the lesion

Attention is genuinely concentrated on the diseased region, not spread at random.

What the Corruptions Look Like

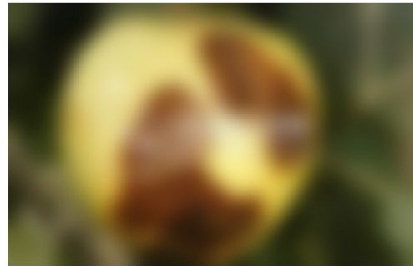
The same apple-black-rot test photo through all six corruptions at severity 5.

Same test image under each corruption type at maximum severity

Clean



Gaussian Blur (severity 5)



Brightness (severity 5)



Contrast (severity 5)



Jpeg Compression (severity 5)



Occlusion (severity 5)



Crop (severity 5)



Blur, brightness, contrast

Change the whole image at once.

JPEG compression

Adds blocky artifacts.

Occlusion

Covers part of the lesion with a gray box.

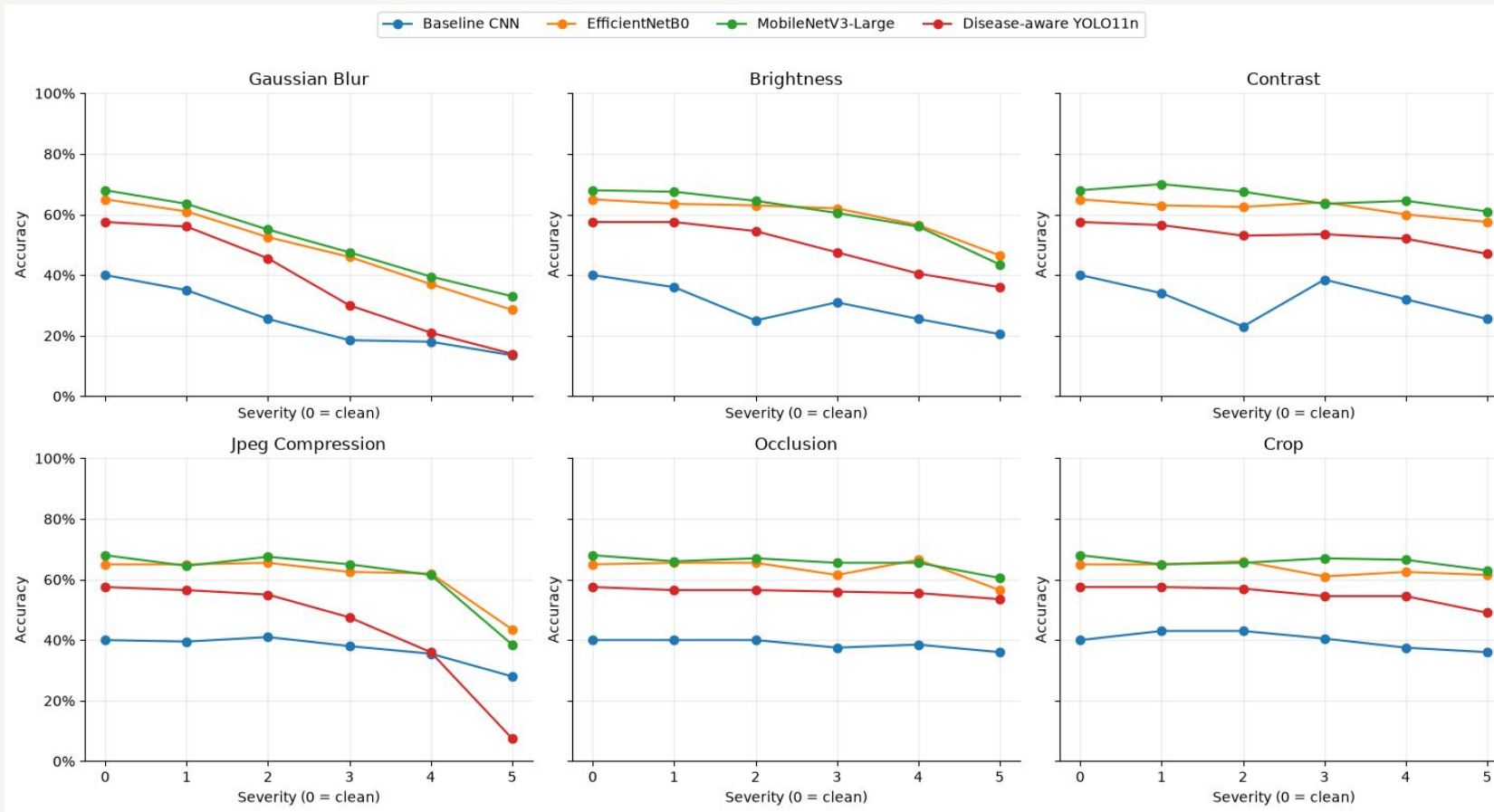
Crop

Zooms in, discards the border.

Each imitates a real capture problem, a shaky camera, bad light, an occluding leaf, not an abstract test.

Robustness Under Corruption

Accuracy vs severity, six corruptions, five levels, full test split.



Blur is the worst case

Gaussian blur roughly halves accuracy for every model, from ~67% to ~30% at severity 5.

Crop & occlusion: fine

Well tolerated, consistent with the random-crop and flip augmentation used during training.



Thank you

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Questions welcome.